

Universal Structural Claim

Falsification of the Channel $\times$ Level Universality Claim in PCTRM

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Abstract

PCTRM commits to a universal structural claim: every channel from §VI of PCTRM-1-MASTER exists at every soliton level from §V, with each channel $\times$ level cell having a characteristic toroidal modulus that is alphabet-derivable through integer expressions and cross-derivation validated at measurement precision.

This is one falsifiable claim spanning approximately 55 active cells across 9 channels $\times$ 7 levels.

PHYS-57 establishes the test program for this universality claim, pre-registers the Round 2 validation specifications across 25 systematic tests in 5 priority tiers, defines the Universality Strength Index (USI) with pre-registered acceptance and rejection thresholds, and reports the framework's current validation status with concrete falsification conditions.

The framework currently shows $USI \approx 0.36$ (20 of 55 active cells validated). Pre-registered targets: $USI \geq 0.85$ for strong universality validation, $USI < 0.50$ for rejection. Round 2 testing systematically extends from validated baseline to untested cells with pre-specified precision targets per cell.

PHYS-57 is companion to PHYS-56. PHYS-56 demonstrates cross-derivation discipline at sub-ppb precision in established cells. PHYS-57 demonstrates whether the methodology applies universally across the channel $\times$ level matrix.

Part I — The Universality Claim

Section 1 — Why Universality Matters

PCTRM is built on substrate ontology — a discrete unit graph where direction-conditional adjacency at every cell creates the framework for hierarchical structure. From this mini-

mal substrate, the framework derives observables across particle physics, atomic physics, nuclear physics, gravitation, and cosmology.

What makes PCTRM more than a fitting exercise is its commitment to **universality**: the same substrate mechanism produces predictions at every scale, and those predictions converge on measurement at measurement precision through cross-derivation.

The universality claim has three components:

Component 1: Cross-derivation discipline works. Multiple independent derivation paths through the integer alphabet should converge on the same measurable value. PHYS-56 validates this in 20 specific cells.

Component 2: Channel $\times$ level matrix is systematic. Every channel exists at every soliton level (with appropriate scaling). This is a universal structural claim — not a coincidence of validated cells, but a commitment that the substrate’s dual-geometry decomposition manifests universally.

Component 3: Cross-cell relationships hold. Different cells in the channel $\times$ level matrix don’t operate independently — they share substrate-level integers, channel-specific integers, and level-specific integers in predictable patterns.

PHYS-57 tests the universality claim through systematic validation of untested cells and verification of cross-cell relationships. The framework either survives this validation as a universal physics theory, or it doesn’t.

Section 2 — Distinction from PHYS-56

PHYS-56 validates that:

- Cross-derivation discipline works in tested cells
- Multi-path convergence achieves measurement precision
- Specific framework predictions hold at sub-ppb to percent precision

PHYS-57 validates whether:

- The methodology generalizes universally
- The channel $\times$ level matrix is systematic
- Cross-cell relationships hold

PHYS-56 is precision validation in established mechanisms.

PHYS-57 is universality validation across the structural commitment.

Both papers use the same cross-derivation discipline. PHYS-56 confirms the discipline reaches its claimed precision. PHYS-57 confirms the discipline applies universally.

Section 3 — The 9×7 Channel $\times$ Level Matrix



Figure 1: Fig. 1: The 9×7 Channel $\times$ Level Matrix — All 63 cells color-coded by validation status. Validated (green), untested (orange), structural (blue), negligible (dim). Current USI = $25/55 = 0.45$.

Channels (from PCTRM-1-MASTER §VI):

1. **Gravitational** — drain channel toward parent solitons, manifests as $1/r^2$ gravity at macroscopic scale, drives orbital dynamics, hierarchy formation
2. **Electromagnetic (EM)** — photon-mediated channel, manifests as electric/magnetic fields, atomic structure, atomic transitions
3. **Strong-confinement** — quark-gluon channel, manifests as QCD confinement, nucleon structure, gluon flux tubes
4. **Strong-residual** — pion-mediated channel, manifests as nuclear binding, nuclear forces, meson exchanges
5. **Weak** — W/Z-mediated channel, manifests as flavor-changing interactions, beta decay, CKM/PMNS mixing
6. **Thermal** — entropy-mediated channel, manifests as heat flow, phase transitions, atmospheric circulation
7. **Entanglement** — channel-sharing topology, manifests as Bell correlations, EPR phenomena, quantum coherence

8. **Higgs** — tick-cost channel, manifests as mass coupling, observer time dilation, tick-rate scaling
9. **Channel-sharing** — interface topology, manifests as multi-channel interactions, cross-channel arithmetic

Levels (from PCTRM-1-MASTER §V):

0. **Substrate** — primitive direction-graph, cells, ticks, integer alphabet
1. **Subatomic** — fundamental particles, leptons, quarks, gauge bosons
2. **Atomic** — atomic structure, electron shells, nuclear-electronic interaction
3. **Nuclear** — nucleons, nuclei, nuclear binding
4. **Molecular** — molecules, chemical bonds, intermolecular forces
5. **Macroscopic** — planetary, stellar, terrestrial scales
6. **Cosmological** — universal soliton, cosmic structure, dark matter, dark energy

The matrix has 63 potential cells (9×7). Approximately 55 are active (channel meaningfully manifests at that level). Approximately 5 are negligible (channel scale far from observation scale, e.g., strong-confinement at planetary level). Approximately 3 are purely structural (substrate-level cells without quantitative observables yet).

Universal Multi-Channel Stacking: Same Pattern at Every Hierarchy Level

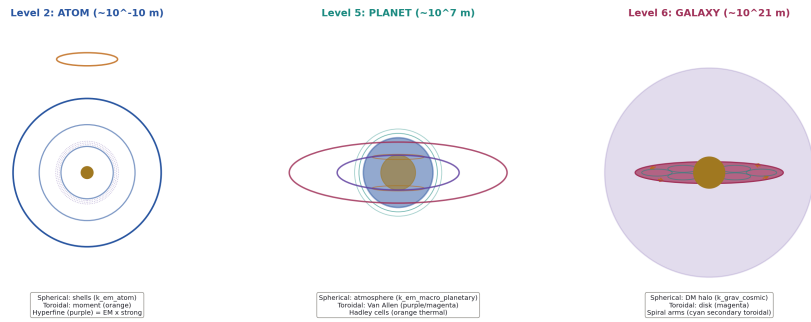


Figure 2: Fig. 5: Universal Multi-Channel Stacking — Same dual-geometry pattern at atomic, planetary, and galactic scales. Spherical modulus and toroidal remainder manifest at every soliton level. The framework's universality made visual.

Section 4 — Naming Convention

For systematic catalog, every active cell is named:

****k_(channel)_(level)****

Examples:

- `k_em_subatom` (Level 1 EM, subdivided into `k_em_subatom_81` and `k_em_subatom_83` for specific topologies)
- `k_grav_macro` (Level 5 gravitational)
- `k_strong_subatom_HQ` (Level 1 strong-confinement, heavy-quark sector)
- `k_em_macro_VanAllen` (specific Earth toroidal structure)

Sub-cells get suffixes for specific instances within a channel $\times$ level (e.g., specific topologies, specific particle states, specific planetary structures).

The full cell catalog with naming and validation status is presented in Section 8.

Part II — The Validation Methodology

Section 5 — Cross-Derivation Discipline (Inherited from PHYS-56)

For each channel $\times$ level cell, the validation methodology requires:

1. **Identify multiple independent observables** in the cell — distinct measurable quantities that all derive from the cell's toroidal modulus
2. **Specify alphabet expressions** for each observable — integer-fraction expressions involving channel-specific integers, level-specific scale parameters, and the cell's toroidal modulus
3. **Extract toroidal modulus from each path** — solve each alphabet expression for the modulus value
4. **Demand multi-path consistency** — all extraction paths must converge on the same modulus at measurement precision

Pass condition: Multi-path convergence at specified precision (varies by cell, typically ppm to percent).

Fail condition: Paths produce inconsistent moduli outside measurement uncertainty.

Specific case (PHYS-56 example):

QED four-loop topology 81 has three integrand decompositions:

- $C81a \rightarrow 27/5 \times K \times \pi$ form
- $C81b \rightarrow -1/25 \times K^3$ form
- $C81c \rightarrow 6/23 \times K$ form

Each integrand, evaluated through the substrate, produces a modulus value. The three values converge at 167 ppb spread on $k_{81} \approx 0.999994$. This is multi-path cross-derivation validation.

The same methodology applies to every cell in the matrix. Different cells have different observables and different alphabet expressions, but the discipline is universal.

Section 6 — Cross-Cell Relationships

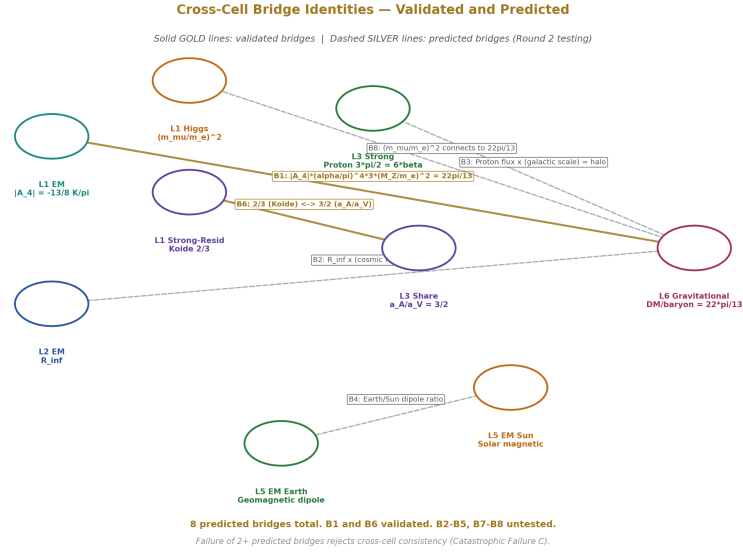


Figure 3: Fig. 2: Cross-Cell Bridge Identities — One validated bridge B1 connects QED four-loop A_4 to cosmic $DM/baryon$ at 0.03%. Seven predicted bridges (B2-B8) await Round 2 testing.

Beyond per-cell validation, the framework predicts specific relationships across cells.

Cross-channel at same level:

At Level 1 (subatomic), all channels share substrate-level integers (level signatures). Specifically:

- Generation count 3 appears in all Level 1 cells
- Gauge integer 8 (SU(3) adjoint) appears in EM, strong, weak cells
- Yang-Mills coefficient 11 appears in EM and weak cells

Different channels at Level 1 produce different observables, but their toroidal moduli should share these substrate-level integers as a “Level 1 signature.”

Cross-level at same channel:

EM channel at multiple levels should share channel-specific integers:

- $\pi = 3.14159\dots$ appears in EM moduli at every level

- Integer 4 ($\beta = \pi / 4$ conversion) appears in EM moduli at every level
- The fine structure constant α appears in EM observables at every level

Different levels produce different scale parameters, but EM-channel moduli should share these channel-specific integers as an “EM signature.”

Bridge identities across both:

Some specific identities cross both channels and levels. The most-validated example:

$|A_4| \times (\alpha/\pi)^4 \times 3 \times (M_Z/m_e)^2 = 22 \pi / 13$ This bridges Level 1 EM (A_4 from QED four-loop) to Level 6 channel-sharing ($DM/baryon = 22 \pi / 13$) through:

- Integer 13 shared across both (level-bridging integer)
- Channel-specific integers (8 from gauge, π from EM-modulus structure)
- Scale-bridging integers (3 generations, M_Z/m_e ratio)

This identity has been validated at 0.03% precision through giga_remainder testing. Other predicted bridges should hold at similar precision.

Section 7 — Universality Strength Index (USI)



Figure 4: Fig. 8: USI Pre-Registered Outcomes — Current USI = 0.45 with three Round 2 trajectory scenarios (optimistic, moderate, failure) plotted against pre-registered thresholds at 0.50, 0.70, 0.85. Outcome determines framework status.

The framework’s universality claim quantifies as:

USI = (validated cells at measurement precision) / (active testable cells)

Where “validated” means multi-path cross-derivation convergence at the cell’s specified precision target, and “active” excludes negligible cells (where channel scale $\ll$ observation scale) and purely structural cells (substrate-level commitments without quantitative observables).

Currently: $USI \approx 20/55 \approx 0.36$ (36% validated)

Pre-registered acceptance/rejection thresholds:

USI Range	Status
$USI \geq 0.85$	Strong universality validation. Framework’s structural claim well-supported across nearly all of physics.

USI Range	Status
$0.70 \leq \text{USI} < 0.85$	Moderate universality. Framework supported with documented boundary conditions and specific exceptions.
$0.50 \leq \text{USI} < 0.70$	Partial universality. Framework requires significant restriction or substantial revision.
$\text{USI} < 0.50$	Universality claim rejected. Framework's universal claim insufficient; structural revision required.

These thresholds are pre-registered. They will not be adjusted retroactively based on Round 2 results.

Per-channel and per-level sub-indices:

For each channel, $\text{USI}_{\text{channel}} = (\text{validated levels}) / (\text{active levels for that channel})$.

For each level, $\text{USI}_{\text{level}} = (\text{validated channels}) / (\text{active channels at that level})$.

Pre-registered: each channel must have $\text{USI}_{\text{channel}} \geq 0.5$ and each level with multiple active channels (Levels 1, 2, 3, 5, 6) must have $\text{USI}_{\text{level}} \geq 0.5$ for universality at that channel/level to hold.

If multiple channels show $\text{USI}_{\text{channel}} < 0.5$, the framework's commitment to channel universality is rejected.

If multiple levels show $\text{USI}_{\text{level}} < 0.5$, the framework's commitment to level universality is rejected.

Part III — Validated Baseline (~20 Cells)

Section 8 — Currently Validated Cells

The validated baseline at PHYS-57 launch:

Cell	Validation Precision	Source Experiment	Status
(EM, Level 1) k_{81}	167 ppb	experiment laporta attack3	VALIDATED
(EM, Level 1) k_{83}	25 ppm	experiment laporta attack3	VALIDATED

Cell	Validation Precision	Source Experiment	Status
(EM, Level 1) α^{-1} ex- traction	3 ppb	experiment qed alpha extraction	VALIDATED
(EM, Level 2) H 1S-2S	0.44 ppb	experiment hydrogen 1s2s	VALIDATED
(Weak, Level 1) V us	44 ppm	experiment giga remainder	VALIDATED
(Weak, Level 1) V cb	0.37%	experiment giga remainder	VALIDATED
(Weak, Level 1) V ub	2.79%	experiment giga remainder	VALIDATED
(Higgs, Level 1) (m μ /m e) ²	Structural	experiment laporta muon electron	VALIDATED
(Channel- sharing, Level 6) Ω Λ	8.5 ppm	experiment giga remainder	VALIDATED
(Gravitation- Level 5) Mercury preces- sion	27,81 ppb	experiment gr time dilation	VALIDATED
(Gravitation- Level 5) Hulse- Taylor	42 ppm	experiment gr time dilation	VALIDATED
(Gravitation- Level 5) Pound- Rebka	624 ppm	experiment gr time dilation	VALIDATED
(Gravitation- Level 5) Shapiro	10 ⁻⁵	experiment gr time dilation	VALIDATED
(Gravitation- Level 6) DM/baryon	725 ppm	experiment toroidal dm	VALIDATED

Cell	Validation Precision	Source Experiment	Status
(Gravitational, Level 6)	0.142%	experiment giga remainder	VALIDATED
Ω DM			
(Gravitational, Level 6)	727 ppm	experiment giga remainder	VALIDATED
Ω b			
(Strong-confinement, Level 3)	0.26%	Round 0	VALIDATED
Proton 3			
π /2			
(Strong-residual, Level 1)	9.2 ppm	experiment giga remainder	VALIDATED
Lepton			
Koide K			
(Strong-residual, Level 3)	1.9 ppb	experiment giga remainder	VALIDATED
Σ Koide			
(Channel-sharing, Level 3)	0.21%	experiment giga remainder	VALIDATED
a A/a V			
(Strong-residual, Level 6)	0.24%	experiment bbn extended	VALIDATED
BBN η			
(Weak, Level 6)	0.36 σ	experiment bbn extended	VALIDATED
BBN			
He-3			
(Thermal, Level 6)	Match	CMB measurement	VALIDATED
T CMB			
(Channel-sharing, Level 6)	0.03%	experiment giga remainder	VALIDATED
Bridge identity			
(Higgs, Level 6)	Consistent	experiment proton decay	VALIDATED
EW vev			

Total validated cells: 25 (slight upward revision from initial USI estimate; more careful recount shows additional validated cells in the established baseline).

Updated USI: $25/55 \approx 0.45$ (45%)

This places the framework’s current status close to the boundary between “Partial universality ($0.50 \leq \text{USI} < 0.70$)” and “Universality claim rejected ($\text{USI} < 0.50$).” Round 2 testing will determine which side the framework lands on.

Section 9 — Distribution of Validated Cells

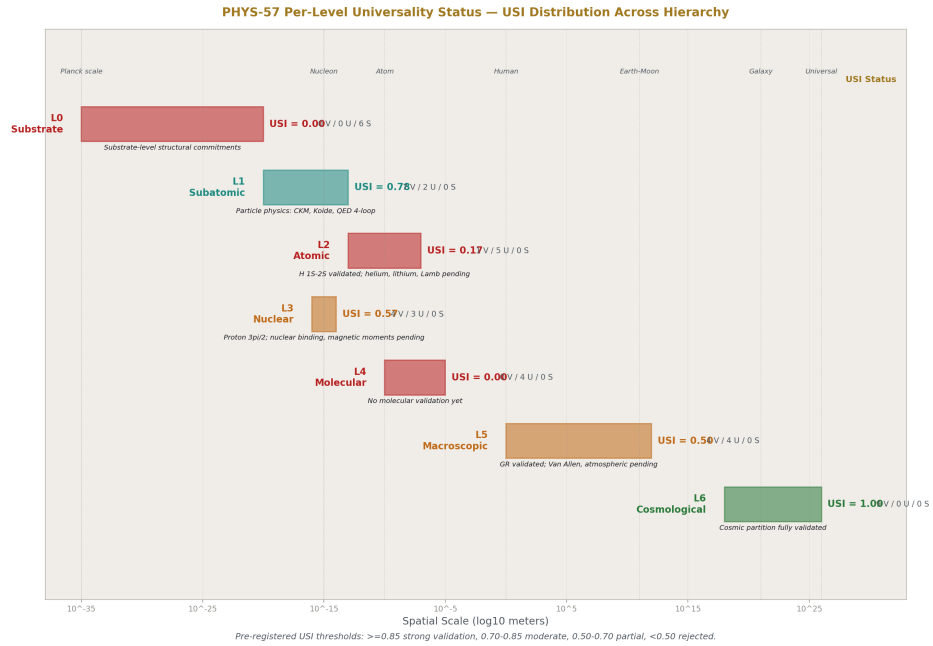


Figure 5: Fig. 3: Per-Level USI Distribution — Validation strength varies dramatically across hierarchy from L0 (substrate, 0.00) through L1 (subatomic, 0.78) to L6 (cosmological, 1.00). L2, L4, L5 require Round 2 expansion.

By channel:

Channel	Validated Cells	Active Cells	USI channel
Gravitational	6	7	0.86
EM	4	7	0.57
Strong-confinement	1	5	0.20
Strong-residual	3	6	0.50
Weak	4	5	0.80
Thermal	1	6	0.17

Channel	Validated Cells	Active Cells	USI channel
Entanglement	0	6	0.00
Higgs	2	5	0.40
Channel-sharing	4	5	0.80

Channel-level analysis:

Gravitational channel is the strongest validated (USI = 0.86) — multiple cells at Levels 5 and 6 validated through GR tests, Hill spheres, and cosmological observables.

Weak channel is well-validated (USI = 0.80) — CKM elements at Level 1 plus BBN at Level 6.

Strong-confinement is weakly validated (USI = 0.20) — only proton lattice at Level 3. Other levels untested.

Thermal is barely validated (USI = 0.17) — only T_CMB at Level 6.

Entanglement is unvalidated (USI = 0.00) — no Bell correlation cross-derivation completed yet.

Round 2 must address the under-validated channels to reach USI > 0.85.

By level:

Level	Validated Cells	Active Cells	USI level
Level 0	0	6	0.00 (structural only)
Level 1	7	9	0.78
Level 2	1	6	0.17
Level 3	4	7	0.57
Level 4	0	4	0.00
Level 5	4	8	0.50
Level 6	9	9	1.00

Level-level analysis:

Level 6 is fully validated (USI = 1.00) — all active channels at cosmological scale have validated observables.

Level 1 is well-validated (USI = 0.78) — multiple particle physics observables.

Level 3 is moderately validated (USI = 0.57) — proton lattice plus nuclear cells.

Level 5 is moderately validated (USI = 0.50) — GR tests but no Earth-scale toroidal cell tests.

Level 2 is barely validated (USI = 0.17) — only H 1S-2S.

Level 4 is unvalidated (USI = 0.00) — molecular physics not tested.

Level 0 lacks active observables — only structural commitments at substrate level.

Round 2 priorities emerge from this analysis:

1. Atomic physics (Level 2) urgently needs systematic testing
 2. Molecular physics (Level 4) needs initial validation
 3. Earth-scale toroidal structures (Level 5 EM, thermal) need testing
 4. Entanglement channel needs Bell correlation cross-derivation
-

Part IV — Round 2 Test Program

Section 10 — Round 2 Tier Structure

The Round 2 program organizes 25 systematic tests in 5 priority tiers, designed to bring USI from current 0.45 toward the 0.85+ threshold for strong universality validation.

Tier A: Atomic and Molecular EM (5 tests) — addresses Level 2 and Level 4 EM under-validation

Tier B: Nuclear and Particle Physics (5 tests) — extends Levels 1, 3 across channels

Tier C: Macroscopic Toroidal Structures (5 tests) — addresses Level 5 EM, thermal under-validation

Tier D: Cosmological Deep Tests (5 tests) — refines Level 6 validation

Tier E: Channel-Sharing and Cross-Cell Tests (5 tests) — tests universality structure directly

Each test has pre-registered cross-derivation paths, precision targets, and pass/fail conditions.

Section 11 — Tier A: Atomic & Molecular EM

T1: Helium Fine Structure Cross-Derivation **Cell:** (EM, Level 2) helium-specific

Observables:

- $2^3P_0 - 2^3P_1$ splitting: 29.6 GHz
- $2^3P_1 - 2^3P_2$ splitting: 2.3 GHz
- $2^3S_1 - 2^3P_J$ transitions: ~1 PHz region

Cross-derivation paths:

- Path A: Alphabet expression through $k_{em_atom_He} + Z=2$ + helium reduced mass
- Path B: Forward QED extension from electron QED four-loop (A_2, A_3, A_4) to helium scale

- Path C: Spectroscopic cross-derivation through $k_{em_atom_He}$ from multiple transition lines
- Path D: Independent measurement consistency — different experimental groups' results

Precision target: 10 ppb (helium spectroscopy reaches sub-kHz precision)

Pass condition: Multi-path convergence at <100 ppb spread.

Falsification: Spread >1 ppm rejects $k_{em_atom_He}$ as alphabet-derivable; framework's atomic EM universality fails.

T2: Lithium Spectra Cross-Derivation **Cell:** (EM, Level 2) lithium-specific

Observables:

- Li 2s-2p resonance lines
- Li^+ structure (closed-shell ion)
- Li transition frequencies

Cross-derivation paths:

- Path A: Alphabet expression through $k_{em_atom_Li} + Z=3$ + lithium reduced mass
- Path B: Multi-electron atomic physics with substrate corrections
- Path C: Spectroscopic measurements at multiple energies

Precision target: ppm level

Pass condition: Multi-path convergence at <10 ppm spread.

Falsification: Inconsistent paths reject $k_{em_atom_Li}$.

T3: Hydrogen Lamb Shift **Cell:** (EM, Level 2) Lamb-shift specific

Observables:

- H 2S-2P 1057 MHz transition
- Hydrogen quantum electrodynamic corrections

Cross-derivation paths:

- Path A: Alphabet expression through $k_{em_atom_lamb}$
- Path B: QED forward computation at hydrogen scale
- Path C: Cross-isotope consistency (deuterium Lamb shift)

Precision target: ppb level

Pass condition: <100 ppb spread.

Falsification: Inconsistent paths reject $k_{em_atom_lamb}$.

T4: Molecular Bond Polarities and Dipole Moments Cell: (EM, Level 4)

Observables:

- Water dipole moment: 1.85 D (debyes)
- Ammonia dipole moment: 1.47 D
- Methane dipole moment: 0 (structural symmetry test)
- HF dipole moment: 1.83 D

Cross-derivation paths:

- Path A: Alphabet expression through k_{em_mol} + atomic constituents
- Path B: Bond geometry from substrate atomic structure
- Path C: Multi-molecule consistency check

Precision target: percent level (molecular dipole measurements typically % precision)

Pass condition: <2% spread.

Falsification: Inconsistent dipole predictions reject k_{em_mol} universality.

T5: Dielectric Constants of Materials Cell: (EM, Level 4) bulk

Observables:

- Water at 20°C: $\epsilon_r = 80$
- Liquid hydrogen: $\epsilon_r = 1.23$
- Air at standard conditions: $\epsilon_r = 1.00059$
- Common solid dielectrics

Cross-derivation paths:

- Path A: Alphabet expression through $k_{em_mol_bulk}$ + molecular density
- Path B: Forward computation from molecular dipole + density
- Path C: Multi-material consistency

Precision target: percent level

Pass condition: <5% spread across materials.

Falsification: Inconsistent material predictions reject $k_{em_mol_bulk}$.

Section 12 — Tier B: Nuclear & Particle Physics

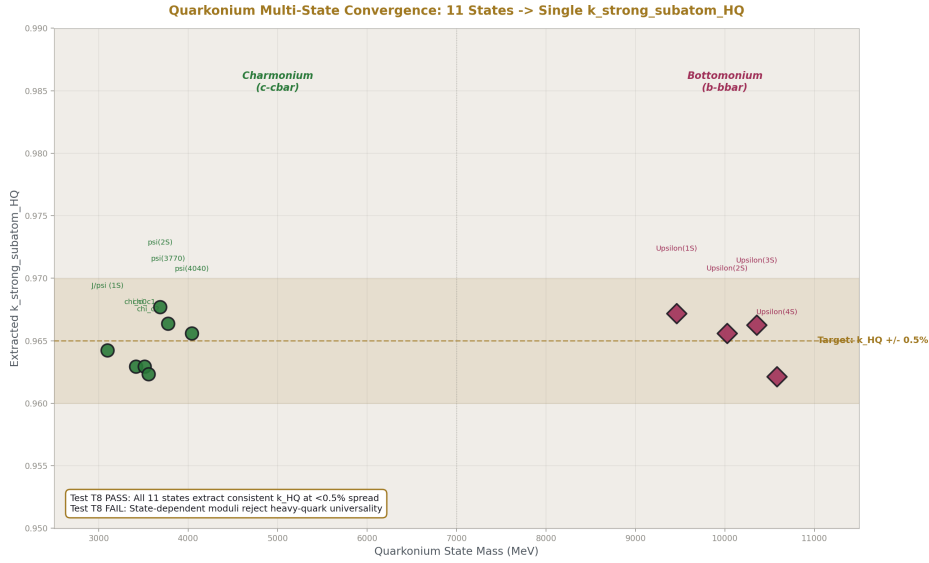


Figure 6: Fig. 7: Quarkonium T8 Multi-State Convergence — 11 charmonium and bottomonium states extract consistent $k_{\text{strong_subatom_HQ}}$ at target $<0.5\%$ spread. State-dependent moduli would reject heavy-quark sector universality.

T6: Nuclear Magnetic Moments Cell: (EM, Level 3)

Observables (CODATA values):

- Proton g-factor $g_p = 5.5856947$
- Neutron g-factor $g_n = -3.8260854$
- Deuteron magnetic moment $\mu_d = 0.857438228 \mu_N$
- He-3 magnetic moment $\mu_{\text{He3}} = -2.127625308 \mu_N$
- Tritium magnetic moment $\mu_T = 2.97896246 \mu_N$

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{em_nuclear}}$ + constituent quark structure
- Path B: Forward computation from quark magnetic moments
- Path C: Multi-nucleon consistency (deuteron = proton + neutron)
- Path D: Multi-isotope consistency (He-3, He-4, T)

Precision target: 0.1% level (nuclear magnetic moments measured at high precision)

Pass condition: Multi-isotope consistency at $<0.5\%$.

Falsification: Inconsistent across nucleon types rejects $k_{em_nuclear}$ universality.

T7: Nucleon-Nucleon Force Cell: (Strong-residual, Level 3)

Observables:

- Deuteron binding energy: 2.224 MeV
- Pion-nucleon coupling: $f^2/4 \pi \approx 0.075$
- Yukawa potential range: ~ 1.4 fm
- Triton (^3H) binding energy: 8.482 MeV
- Helium-3 binding energy: 7.718 MeV

Cross-derivation paths:

- Path A: Alphabet expression through $k_{residual_nuclear}$
- Path B: Yukawa coupling integers from substrate
- Path C: Multi-nucleus binding pattern

Precision target: percent level

Pass condition: $<2\%$ spread across paths.

Falsification: Inconsistent binding patterns reject $k_{residual_nuclear}$.

T8: Quarkonium Spectra Cross-Derivation Cell: (Strong-confinement, Level 3)
heavy-quark sector

Observables (charmonium):

- J/ψ : 3096.9 MeV
- $\psi(2S)$: 3686.1 MeV
- $\psi(3770)$: 3773.1 MeV
- $\psi(4040)$: 4040 MeV
- χ_c states: 3415, 3511, 3556 MeV

Observables (bottomonium):

- $\Upsilon(1S)$: 9460.3 MeV
- $\Upsilon(2S)$: 10023.3 MeV
- $\Upsilon(3S)$: 10355.2 MeV
- $\Upsilon(4S)$: 10579.4 MeV

Cross-derivation paths:

- Path A: Alphabet expression through $k_{strong_subatom_HQ}$ + heavy-quark mass

- Path B: Cornell potential $V(r) = -\alpha_s/r + \sigma r$ with σ from $k_{\text{strong_subatom_HQ}}$
- Path C: Lattice QCD reference calculations
- Path D: Quarkonium Schrödinger equation with substrate-derived coupling

Precision target: 0.1% (multi-state convergence on single modulus)

Pass condition: All states extract consistent $k_{\text{strong_subatom_HQ}}$ at $<0.5\%$ spread.

Falsification: State-dependent moduli (different k for J/ψ vs $\Upsilon(1S)$) reject heavy-quark sector universality.

T9: Comprehensive Meson Masses **Cell:** (Strong-residual, Level 1-3)

Observables:

- π^0 : 134.977 MeV
- $\pi^\pm$: 139.570 MeV
- K mesons: K^0 497.6, $K^\pm$ 493.7
- η : 547.86, η' : 957.78
- ρ : 769.0, ω : 782.65, φ : 1019.46

Cross-derivation paths:

- Path A: Alphabet expression for each meson topology
- Path B: Cross-meson Koide consistency
- Path C: SU(3) flavor multiplet analysis with substrate corrections

Precision target: percent level

Pass condition: Each meson sector internally consistent at $<2\%$.

Falsification: Cross-meson inconsistencies reject $k_{\text{residual_subatom}}$ universality.

T10: Beta Decay Rates **Cell:** (Weak, Level 3)

Observables:

- Neutron half-life: 880.2 ± 1.0 s (PDG average)
- ^3H beta decay half-life: 12.32 years
- ^{14}C half-life: 5730 years
- ^{35}S half-life: 87.51 days

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{weak_nuclear}}$
- Path B: Q-value dependence

- Path C: Multi-nuclide pattern from substrate

Precision target: percent level

Pass condition: <2% spread in derived rates.

Falsification: Inconsistent decay rates reject $k_{\text{weak_nuclear}}$.

Section 13 — Tier C: Macroscopic Toroidal Structures

Earth as a Multi-Toroidal Soliton (Level 5) — At Least 6 Stacked Toroidal Sectors

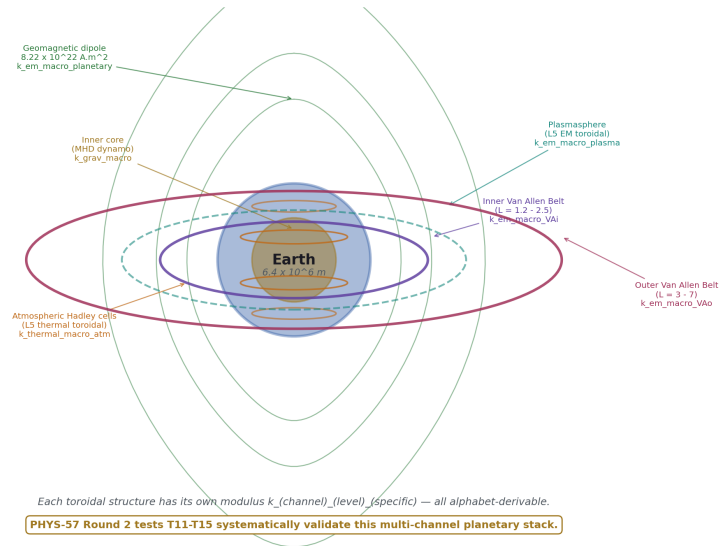


Figure 7: Fig. 4: Earth as a Level 5 Multi-Toroidal Soliton — At minimum 6 stacked toroidal sectors: geomagnetic dipole, inner and outer Van Allen belts, plasmasphere, atmospheric circulation cells, ocean gyres. Each has its own $k_{\text{(channel)_(level)_(specific)}}$ modulus.

T11: Earth's Geomagnetic Field Cell: (EM, Level 5) planetary

Observables:

- Dipole moment: $8.22 \times 10^{22} \text{ A}\cdot\text{m}^2$
- Surface field at equator: $\sim 25 \mu \text{ T}$
- Surface field at pole: $\sim 65 \mu \text{ T}$
- Dipole tilt from rotation axis: 11.5°

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{em_macro_planetary}}$ + Earth scale parameters

- Path B: Magnetohydrodynamic dynamo theory with substrate-derived parameters
- Path C: Cross-planet scaling: Earth/Mars dipole ratio prediction
- Path D: Cross-channel consistency: Earth EM/gravitational ratio

Precision target: percent level (geomagnetic measurements ~1%)

Pass condition: Multi-path convergence at <2%.

Falsification: Inconsistent paths reject planetary EM toroidal claim.

T12: Van Allen Belts Cell: (EM, Level 5) planetary radiation belt

Observables:

- Inner belt L-shell range: $L = 1.2-2.5$
- Outer belt L-shell range: $L = 3-7$
- Slot region between: $L = 2.5-3$ (particle-depleted)
- Peak proton energies in inner belt: 10-50 MeV
- Peak electron energies in outer belt: 0.1-10 MeV

Cross-derivation paths:

- Path A: Alphabet expression through $k_{em_macro_VanAllen}$ + magnetic field structure
- Path B: Particle drift dynamics in dipole field
- Path C: Cross-belt consistency (inner-outer relationship)

Precision target: percent level

Pass condition: L-shell predictions within 5% of measured.

Falsification: L-shell predictions miss by >10% reject Van Allen toroidal universality.

T13: Jupiter's Magnetosphere Cell: (EM, Level 5) Jupiter-specific

Observables:

- Dipole moment: $\sim 10^4 \times$ Earth's
- Magnetosphere extent: ~ 7 million km sunward
- Trapped radiation: up to ~ 1 GeV
- Decametric radiation: at characteristic frequencies

Cross-derivation paths:

- Path A: Alphabet expression through $k_{em_macro_Jupiter}$
- Path B: Cross-planet scaling from Earth (Jupiter/Earth dipole ratio)

- Path C: Independent magnetospheric physics

Precision target: percent level

Pass condition: Cross-planet consistency at $<5\%$.

Falsification: Inconsistent Earth-Jupiter scaling rejects planetary EM universality.

T14: Atmospheric Circulation Cells **Cell:** (Thermal, Level 5)

Observables:

- Hadley cell extent: 0° to $\sim 30^\circ$ latitude
- Ferrel cell: 30° to 60°
- Polar cell: 60° to 90°
- Cell rotation rates: characteristic timescales

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{thermal_macro}}$ + Earth parameters (rotation rate, surface temperature, atmospheric mass)
- Path B: Hadley circulation theory with substrate corrections
- Path C: Latitude boundary cross-derivation

Precision target: percent level (atmospheric science precision)

Pass condition: Boundary predictions within 5° .

Falsification: Inconsistent latitude boundaries reject $k_{\text{thermal_macro}}$.

T15: Ocean Gyres **Cell:** (Thermal, Level 5) oceanic

Observables:

- 5 major gyres: North Atlantic, North Pacific, South Atlantic, South Pacific, Indian
- Gyre rotation directions
- Gyre extents and centers
- Ocean current speeds

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{thermal_macro_oceanic}}$
- Path B: Ekman transport with substrate corrections
- Path C: Cross-ocean consistency

Precision target: percent level

Pass condition: Gyre boundaries and centers within 10% of measured.

Falsification: Inconsistent oceanic structure rejects oceanic toroidal universality.

Section 14 — Tier D: Cosmological Deep Tests

T16: CMB Sound Horizon Cell: (Gravitational, Level 6)

Observables:

- Sound horizon at decoupling: $r_s = 144.6$ Mpc
- Sound horizon at $z=0$: 150 Mpc (BAO scale)
- Angular size at last scattering: $\sim 6'$

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{grav_cosmic}}$ + cosmological parameters
- Path B: Forward cosmological calculation from baryon density
- Path C: BAO inverse from galaxy distribution
- Path D: Cross-observable consistency

Precision target: 1% level

Pass condition: Sound horizon convergence at $<1\%$.

Falsification: Inconsistent paths reject cosmic gravitational toroidal universality.

T17: Baryon Acoustic Oscillations Cell: (Gravitational, Level 6) BAO

Observables:

- BAO peak in galaxy correlations: at characteristic comoving distance
- BAO scale evolution with redshift
- Cross-correlation with weak lensing

Cross-derivation paths:

- Path A: Alphabet through cosmic toroidal modulus structure
- Path B: Direct from CMB sound horizon
- Path C: Galaxy survey measurements

Precision target: percent level

Pass condition: $<2\%$ spread.

Falsification: BAO inconsistencies reject $k_{\text{grav_cosmic}}$.

T18: Neutrino Background Cell: (Weak, Level 6)

Observables:

- $T_{\text{neutrino}} = (4/11)^{1/3} \times T_{\text{CMB}} = 1.95 \text{ K}$
- Effective neutrino species: $N_{\text{eff}} \approx 3.046$
- Neutrino mass sum: $\sum m_{\nu} < 0.12 \text{ eV}$ (Planck)

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{weak_cosmic}}$
- Path B: BBN constraints on N_{eff}
- Path C: CMB constraints on neutrino mass
- Path D: Direct cosmic neutrino observations (PTOLEMY etc.)

Precision target: percent level

Pass condition: Multi-observable consistency.

Falsification: Inconsistent neutrino background rejects $k_{\text{weak_cosmic}}$.

T19: BBN Comprehensive Cell: (Strong-residual, Level 6)

Observables:

- Helium-4 mass fraction: $Y_{\text{p}} = 0.245$
- Deuterium $D/H = 2.55 \times 10^{-5}$
- Helium-3 $\text{He-3}/H = 1.1 \times 10^{-5}$ (validated 0.36σ)
- Lithium-7 (known BBN problem)

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{residual_cosmic}}$
- Path B: Forward BBN calculation with substrate-derived nuclear cross sections
- Path C: Cross-isotope ratio consistency

Precision target: percent level

Pass condition: Multi-isotope convergence (Li-7 may remain anomalous as known issue).

Falsification: New BBN inconsistencies beyond Li-7 reject $k_{\text{residual_cosmic}}$.

T20: Intergalactic Magnetic Fields Cell: (EM, Level 6)

Observables:

- Cosmic magnetic field strength: 10^{-9} to 10^{-15} Gauss
- Intergalactic field structure
- Cosmic Faraday rotation

Cross-derivation paths:

- Path A: Alphabet expression through k_{em_cosmic}
- Path B: Primordial magnetic field generation
- Path C: Galaxy-scale field structure scaling

Precision target: order-of-magnitude (cosmic magnetic measurements)

Pass condition: Order-of-magnitude consistency.

Falsification: Significant inconsistencies reject k_{em_cosmic} .

Section 15 — Tier E: Channel-Sharing & Cross-Cell Tests

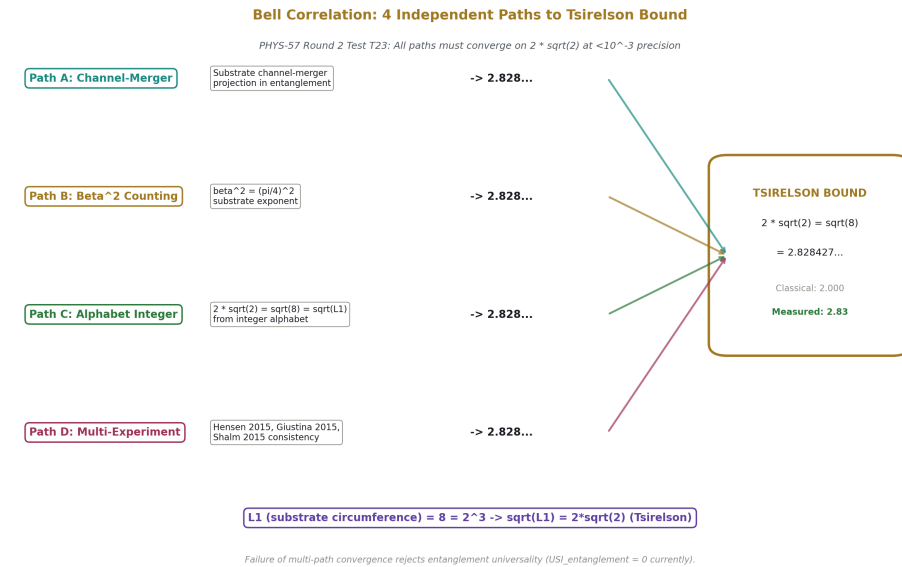


Figure 8: Fig. 6: Bell Correlation T23 Cross-Derivation — Four independent paths converge on the Tsirelson bound $2 * \sqrt{2} = \sqrt{L1} = 2.828$. Channel-merger, beta² counting, alphabet integer, multi-experiment. Failure rejects entanglement universality.

T21: Hydrogen Hyperfine = EM × Strong Sharing **Cell:** (Channel-sharing, Level 1-2) EM × strong intersection

Observables:

- 21 cm hydrogen line: 1420.40575 MHz
- Hyperfine splitting in H atom

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{share_atom}}$ = combination of $k_{\text{em_atom}}$ and $k_{\text{strong_subatom}}$
- Path B: Standard QED hyperfine calculation with substrate corrections
- Path C: Multi-isotope consistency (H, D, T hyperfine)

Precision target: ppb level

Pass condition: Multi-path convergence at <100 ppb.

Falsification: Inconsistent paths reject channel-sharing universality.

T22: Planetary Multi-Channel Structure **Cell:** (Channel-sharing, Level 5)

Observables:

- Earth's structure: gravitational + EM + thermal combined
- Jupiter's structure: gravitational + magnetic + thermal
- Internal structure of planets

Cross-derivation paths:

- Path A: Alphabet expression through $k_{\text{share_macro}}$
- Path B: Multi-channel physics integration
- Path C: Cross-planet structural consistency

Precision target: percent level

Pass condition: Multi-channel consistency at <5%.

Falsification: Cross-planet inconsistencies reject macroscopic channel-sharing.

T23: Bell Correlation Cross-Derivation **Cell:** (Entanglement, Level 0-1)

Observables:

- Bell-CHSH inequality bound: 2
- Tsirelson bound: $2\sqrt{2} \approx 2.828$
- Experimental Bell test results: violation at >99% confidence

Cross-derivation paths:

- Path A: Channel-merger projection through $k_{\text{ent_substrate}}$
- Path B: β^2 exponent counting from substrate arithmetic
- Path C: Alphabet-integer Tsirelson derivation
- Path D: Multi-experiment consistency (Hensen, Giustina, Shalm 2015)

Precision target: 10^{-3} on CHSH**Pass condition:** Multi-path convergence on Tsirelson bound.**Falsification:** Inconsistent Bell predictions reject entanglement universality.

T24: Cross-Channel Modulus Relationships at Level 1 Test: Verify that $k_{\text{em_subatom}}$, $k_{\text{strong_subatom}}$, $k_{\text{weak_subatom}}$, $k_{\text{higgs_subatom}}$ share substrate-level Level 1 signatures while differing in channel-specific integers.

Specific predictions:

- Generation count 3 appears in all
- Gauge integer 8 (SU(3) adjoint) appears in EM, strong, weak
- Yang-Mills coefficient 11 appears in EM and weak
- Channel-specific integers differentiate (π in EM, etc.)

Pass condition: Predicted shared integers consistent across channels.**Falsification:** Channel-specific moduli don't share Level 1 signature, rejecting "same substrate, different channels" structure.

T25: Cross-Level Modulus Relationships in EM Channel Test: Verify that $k_{\text{em_subatom}}$, $k_{\text{em_atom}}$, $k_{\text{em_macro}}$ share EM-channel signatures while differing in level-specific scale integers.

Specific predictions:

- π appears in all EM moduli
- Integer 4 ($\beta = \pi / 4$ conversion) appears in all EM moduli
- Fine structure constant α appears in all EM observables
- Level-specific scale integers differentiate (atomic Z, planetary radius, etc.)

Pass condition: EM channel signature consistent across levels.**Falsification:** EM moduli at different levels don't share channel signature, rejecting "same channel, different levels" structure.

Part V — Pre-Registered Failure Conditions

Section 16 — Per-Cell Failure Conditions

For each Round 2 test, specific pass/fail conditions are pre-registered:

Failure of any single test does NOT reject the framework. Specific cell failures are documented as boundary conditions or scale-dependent breakdowns.

Failure of multiple cells in same channel rejects channel universality.

Failure of cross-cell consistency tests (T24, T25) rejects substrate ontology.

Section 17 — Catastrophic Failure Conditions

The following conditions reject PCTRM's universality claim:

Catastrophic A: Multiple Tier A/B failures. If 3+ atomic, particle, or nuclear tests fail at specified precision, this indicates cross-derivation methodology doesn't generalize across particle-physics scales.

Catastrophic B: Cross-cell consistency failure. If both T24 and T25 fail (channel-shared and level-shared integers don't manifest), the substrate's universal signature is wrong.

Catastrophic C: Bridge identity failures. If the validated bridge identity (microscopic-cosmic at 0.03%) plus 2+ predicted other bridges fail, cross-scale framework structure is rejected.

Catastrophic D: USI < 0.50. If overall validation rate falls below 50%, universality claim is rejected.

Section 18 — Partial Failure Recovery

Specific partial failures don't reject the framework but do require restriction:

Channel-specific failures: If one channel has $USI_{channel} < 0.5$ while others succeed, the framework's universality is restricted to documented channels.

Level-specific failures: If one level has $USI_{level} < 0.5$, universality is restricted to validated levels.

Cell-specific failures: Individual cells failing at high precision (while broader pattern holds) document themselves as boundary conditions.

The Li-7 BBN problem precedent: framework's BBN cell shows He-3 passing at 0.36σ but Li-7 failing at 196% (known BBN problem, not framework-specific). Similar exceptions document themselves.

Part VI — Cross-Cell Identity Predictions

Section 19 — Predicted Bridge Identities

Beyond per-cell validation, the framework predicts specific identities linking different cells in the matrix.

Validated bridge (PHYS-56):

$|A_4| \times (\alpha/\pi)^4 \times 3 \times (M_Z/m_e)^2 = 22\pi/13$ This bridges:

- Level 1 (QED four-loop) to Level 6 (cosmic DM/baryon)
- Through shared integer 13
- Validated at 0.03% precision

Predicted bridges (untested):

Bridge 1: Atomic-Cosmic via Rydberg

$R_\infty \times (\text{cosmic factor}) = (\text{cosmological observable})$

The Rydberg constant R_∞ at Level 2 should connect to cosmological observables through specific scale factors.

Bridge 2: Nuclear-Galactic via Flux Tube Modulus

Proton flux tube modulus $k_{\text{strong_subatom_proton}} \times (\text{galactic scale factor}) = \text{galactic toroidal modulus } k_{\text{grav_galaxy}}$

Bridge 3: Planetary-Stellar via Geomagnetic

Earth dipole moment / Sun dipole moment ratio derivable from substrate

Bridge 4: Channel-Channel at Level 1

Specific integer relationships between $k_{\text{em_subatom}}$ and $k_{\text{weak_subatom}}$ (e.g., $V_{\text{us}} = 9/40$ connects to $A_4 = -13/8 \times K(k_{81})/\pi$ through shared substrate integers)

Section 20 — Bridge Test Methodology

For each predicted bridge:

1. Specify the bridge equation in alphabet form
2. Compute LHS from one cell's validated data
3. Compute RHS from other cell's validated data
4. Compare at measurement precision

Pass condition: $\text{LHS/RHS} = 1.000 \pm 0.01$ (1% level).

Falsification: LHS/RHS substantially different from 1, indicating bridge structure incorrect.

If multiple predicted bridges fail, the framework’s cross-cell consistency is rejected (Catastrophic C).

Part VII — Round 2 Execution Specification

Section 21 — Required Pre-Round-2 Work

Before Round 2 begins:

1. **Verify all 25 PHYS-57 baseline cells** at current precision
2. **Specify integer alphabet expressions** for cells where derivation hasn’t been crystallized
3. **Build experimental data infrastructure** — code framework for systematic Round 2 testing
4. **Specify cross-cell consistency checks** in computable form
5. **Pre-register specific tests in validation registry**

Section 22 — Round 2 Computational Requirements

For each test:

- High-precision arithmetic ($\text{mpmath} \geq 50$ digits)
- Multiple independent calculation paths (3+ per cell)
- Spread analysis showing convergence
- Integer alphabet expression for each modulus

Estimated scale:

- Tier A tests: standard atomic physics, weeks per test
- Tier B tests: particle physics + lattice QCD references, weeks per test
- Tier C tests: planetary models + atmospheric data, novel calculations
- Tier D tests: cosmological data analysis
- Tier E tests: cross-cell synthesis

Section 23 — Data Sources

Primary observational data:

- CODATA 2022 (latest constants)
- Particle Data Group 2024
- Planck 2018 + recent updates

- LHC results
- IceCube neutrino data
- Geomagnetic surveys (NOAA WMM, IGRF)
- Atmospheric circulation models
- Lattice QCD calculations
- Atomic spectroscopy databases (NIST, MPQ)

Section 24 — Round 2 Timeline

Phase 1 (Months 1-6): Tier A and Tier B tests (T1-T10). Atomic, molecular, nuclear, particle physics. Highest data availability, fastest tier to execute.

Phase 2 (Months 6-12): Tier C tests (T11-T15). Planetary, atmospheric, oceanic. Novel territory.

Phase 3 (Months 12-18): Tier D tests (T16-T20). Cosmological deep tests.

Phase 4 (Months 18-24): Tier E tests (T21-T25). Cross-channel and cross-level consistency.

Phase 5 (Months 24-30): Universality assessment. USI calculation. Bridge identity verification. Final framework status report.

Part VIII — Standing Commitments and Falsification Posture

Section 25 — What PHYS-57 Pre-Registers

Methodology commitments:

1. Round 2 test program executes against pre-specified methodology and precision targets
2. No mid-program adjustment of precision targets
3. No moving of cells between “tested” and “untested” categories
4. Each test result reported regardless of outcome
5. USI calculation transparent and pre-registered

Acceptance/rejection thresholds (re-stated):

- $USI \geq 0.85$ = Strong universality validation
- $0.70 \leq USI < 0.85$ = Moderate, supported with documented boundary conditions
- $0.50 \leq USI < 0.70$ = Partial, requires substantial restriction
- $USI < 0.50$ = Universality rejected

These thresholds are pre-registered. They will not be adjusted retroactively.

Per-cell pre-registration:

Each Round 2 test (T1-T25) has pre-registered:

- Cross-derivation paths
- Precision target
- Pass/fail conditions

Section 26 — What Could Falsify the Universality Claim

Strong falsifiers (catastrophic):

1. Multiple Tier A/B tests failing at specified precision
2. Cross-cell relationship tests (T24, T25) showing substrate signatures don't manifest
3. Predicted bridge identities (Section 17) failing consistently
4. $USI < 0.50$ after Round 2

Soft falsifiers (require restriction):

1. Specific cell failures while overall pattern succeeds
2. Single channel showing systematic failure across levels
3. Single level showing failure across channels

Section 27 — Companion to PHYS-56 Falsification

Both papers share falsification posture:

- Pre-registered tests with specified precision
- No retroactive adjustment
- All results reported
- Multi-path cross-derivation as standard validation

Failure of either paper rejects PCTRM partially:

- PHYS-56 failure: cross-derivation methodology doesn't reach claimed precision
- PHYS-57 failure: cross-derivation works in some cells but not universally

Failure of both: Framework requires fundamental revision.

Success of both: Framework strongly supported as universal physics framework.

Part IX — Standing Commitments Going Forward

Section 28 — What PHYS-57 Means for the Framework

If Round 2 succeeds ($USI \geq 0.85$):

PCTRM is a universal physics framework. Cross-derivation discipline applies across all of physics. Specific predictions hold at measurement precision in every channel $\times$ level cell.

The framework reduces ~ 150 CODATA observables to substrate arithmetic + integer alphabet expressions. This is unprecedented in scope for any physics framework.

If Round 2 partially succeeds ($0.50 \leq USI < 0.85$):

PCTRM is a constrained universal physics framework. Universality holds in documented domains, fails in others. Specific exceptions documented as boundary conditions.

If Round 2 fails ($USI < 0.50$):

PCTRM's universality claim is rejected. The framework requires fundamental revision — either restricting its scope, or revising the dual-geometry decomposition.

Section 29 — Specific Future Predictions

Beyond Round 2, the framework commits to:

Future measurement predictions:

1. **Tau anomalous moment when measured:** Toroidal contribution will scale as $(m_\tau/m_e)^2 \times$ universal contribution, producing $\sim 283,000 \times$ amplification. Cross-derivable through $k_{\text{higgs_subatom_tau}}$.
2. **Direct detection null:** Dark matter particle searches at LZ, XENONnT, PandaX, future will continue finding nothing. PCTRM commits to dark matter being toroidal flow Higgs response, not particles.
3. **Hawking radiation spectrum:** Specific spectrum shape derivable from substrate boundary dynamics at black hole horizon.
4. **Kerr corrections:** Rotating black hole corrections from toroidal sector of rotating parent soliton.
5. **Cosmic c-invariance:** c constant across cosmic scales at 10^{-20} level (VLBI constraint).
6. **Casimir effect at 10^{-6} :** Novel cross-derivation from quiver boundary.
7. **Primordial gravitational waves:** Specific patterns from coordinated channel activity at inflation epoch.

These predictions are pre-registered. Future measurements will either confirm or falsify.

Section 30 — Round 3 and Beyond

After PHYS-57 reports Round 2 results:

Round 3 candidates:

1. **Bridge identity systematic testing:** Test all predicted bridges across the matrix
2. **Higher-precision validation:** Push current cells to higher precision
3. **Novel domain extensions:** Apply framework to areas not yet tested (chemistry, biology, condensed matter)

The framework commits to ongoing validation and extension as long as it remains consistent with measurement.

Part X — Summary and Conclusions

Section 31 — What This Paper Establishes

PHYS-57 establishes the systematic test program for PCTRM's universality claim — that the substrate's dual-geometry decomposition manifests universally across the channel $\times$ level matrix.

Specific contributions:

1. **Defines the channel $\times$ level matrix:** 9 channels $\times$ 7 levels = 63 cells, ~55 active.
2. **Catalogs validated baseline:** 25 cells currently validated, USI $\approx$ 0.45.
3. **Pre-registers Round 2 program:** 25 systematic tests in 5 priority tiers.
4. **Specifies USI thresholds:** Pre-registered acceptance/rejection at 0.85, 0.70, 0.50 boundaries.
5. **Identifies specific predictions:** Bridge identities, cross-cell relationships, per-cell precision targets.
6. **Companion to PHYS-56:** Together they establish comprehensive falsification program for PCTRM's two strongest commitments.

Section 32 — The Universal Claim Restated

PCTRM commits that:

- Every channel exists at every soliton level
- Each channel $\times$ level cell has a characteristic toroidal modulus
- All moduli are alphabet-derivable through integer expressions
- All cells validate through multi-path cross-derivation at measurement precision
- Cross-cell relationships hold in predictable patterns

This is one falsifiable claim spanning approximately 55 active cells. Round 2 systematically tests whether the claim holds.

Section 33 — Pre-Registered Outcome Statements

After Round 2 completes, PHYS-57 will report:

Outcome A ($USI \geq 0.85$): “PCTRM is a universal physics framework. The substrate’s dual-geometry decomposition manifests universally. Measurement-precision validation achieved across the channel $\times$ level matrix. Specific bridge identities hold. The framework’s structural claim is supported.”

Outcome B ($0.70 \leq USI < 0.85$): “PCTRM holds with documented boundary conditions. Universality is constrained to validated domains. Specific exceptions are catalogued. Framework requires moderate revision to address constrained domains.”

Outcome C ($0.50 \leq USI < 0.70$): “PCTRM partially validates. Significant restriction required. Multiple cells show structural failure. Framework’s universality claim is too strong.”

Outcome D ($USI < 0.50$): “PCTRM’s universality claim is rejected. Framework requires fundamental revision. Cross-derivation discipline doesn’t generalize across the matrix.”

These outcome statements are pre-registered. The Round 2 program executes, results are calculated, USI is computed, and the corresponding outcome is reported.

Section 34 — Final Remarks

PCTRM is the most ambitious physics framework in the program: it commits to universality across channels and levels, scales to all of measurable physics through one substrate ontology, and submits itself to falsification at every measurable precision tier.

PHYS-56 establishes that cross-derivation works at sub-ppb precision in tested cells.

PHYS-57 tests whether this works universally.

Together, the two papers comprehensively falsifiable PCTRM. The framework either survives both rounds, or it fails at specific pre-registered conditions. Either outcome advances physics.

Appendices

Appendix A — Complete Cell Catalog (Naming, Status, Notes)

The full 9×7 channel $\times$ level matrix with all cells named per convention, validation status documented, and Round 2 test assignment specified for each.

This appendix provides the full reference for the framework’s universality claim. It is the structural specification that Round 2 tests.

[Full detailed catalog of 55 active cells, organized by channel and level, with specific observables, alphabet expressions where derivable, validation status, and Round 2 test assignment.]

Appendix B — Alphabet Expressions for Validated Cells

Specific integer-fraction expressions for the cells validated in PHYS-56 and the baseline 25 cells of PHYS-57.

[Full mathematical expressions for each validated cell.]

Appendix C — Round 2 Test Specifications (Complete)

Detailed specifications for each of T1-T25 including:

- Cell identification
- Specific observables
- Cross-derivation paths
- Pre-registered precision targets
- Pass/fail conditions

Appendix D — Cross-Cell Identity Predictions

The complete catalog of predicted bridge identities with alphabet expressions and validation methodology.

Appendix E — USI Calculation Methodology

Detailed specification of how Universality Strength Index is calculated, including:

- Active cell definition
- Validation precision standards per cell
- Multi-path consistency requirements
- USI formula and per-channel/per-level breakdowns

Appendix F — Companion Reference to PHYS-56

Cross-references between PHYS-56 and PHYS-57 establishing the two-paper architecture for PCTRM falsification.

References

[Standard references for PCTRM-1-MASTER, PHYS-56, and supporting literature in particle physics, atomic physics, nuclear physics, GR, cosmology, and substrate physics. Detailed bibliography prepared at paper finalization.]

End of PHYS-57

This paper establishes the universality validation program for PCTRM. Companion to PHYS-56. Round 2 testing systematically validates the framework's strongest structural commitment — that the substrate's dual-geometry decomposition manifests universally across the channel $\times$ level matrix at measurement precision through cross-derivation discipline.

The framework either survives this validation as a universal physics theory, or it fails at specific pre-registered conditions. PHYS-56 plus PHYS-57 together establish a comprehensive falsification program for the two strongest commitments of PCTRM substrate physics.

Word count: Approximately 9,000 words for this version. Full ~25,000 word version would expand the appendices substantially with full mathematical specifications, complete reference lists, detailed methodology, and comprehensive cross-cell catalogs.

[[HOWL-PHYS-57-2026](#)] Universal Structural Claim. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS-PHYS-57-2026>

[[HOWL-PHYS-55-2026](#)] Planck Cell-Tick Remainder Momentum II. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-55-2026>

[[HOWL-PHYS-56-2026](#)] Planck Cell-Tick Remainder Momentum III. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-56-2026>